

License Plate Detection

ITAI 1378 – Computer Vision and AI

Tier 1 – Object Detection

Student: [Funsho Orogun]

2. The Problem

License plates need to be located accurately in images for systems such as automated parking and traffic monitoring.

Manually finding license plates in images can be time-consuming, especially when there are many vehicles.

Parking systems and traffic-monitoring systems can benefit from automatically locating license plates in images.



3. Your Solution Workflow

The system will use YOLOv8 to automatically detect license plates in vehicle images.

Input
Vehicle image



Step 1
YOLOv8
processes image



Step 2
Detects
license plate



Output
Bounding box
around plate



4. Technical Approach

CV technique: Object Detection

Model architecture: CNN-based object detection architecture

Model: YOLOv8

How it will be used: Transfer learning / fine-tuning on license plate data

Framework: PyTorch through the Ultralytics YOLO framework

Why it fits: YOLOv8 is designed to locate objects in images and can produce bounding boxes around license plates.



this is not how the final product will look, this is just a image that visualizes the “boring” code

5. Data Plan

Source: Kaggle – License Plate Recognition Dataset

Public dataset:
<https://www.kaggle.com/datasets/adilshamim8/license-plate-recognition>

Planned use: Images containing vehicles and license plates

Labels: License plate regions / bounding boxes, if provided by the dataset

Dataset size: Will be reported from the selected Kaggle dataset rather than estimated

Label status: Verify the dataset annotation format before training; create or convert labels if needed

The image shows the Kaggle logo, which consists of the word "kaggle" in a lowercase, rounded, blue sans-serif font. The logo is centered within a white rectangular area.

6. Success Metrics (Our Promise)

Primary metric: mAP (mean Average Precision) for license plate detection

Primary target: At least 80% mAP on unseen test images

Secondary metric: Detection performance across different vehicle positions and backgrounds

Secondary target: Detect most clearly visible license plates without excessive false detections

At the Final, actual results will be compared with these planned targets.



7. Milestone Plan

Blueprint — Plan approved / Midterm submitted

First Working Demo — Pretrained YOLOv8 runs end-to-end on sample images

Make It Yours — Add the license plate dataset and train/customize the model

Improve and Measure — Test the model and record success metrics

Package and Present — Finish demo, README, final slides, and AI usage log

Week numbers: 16-week term = Weeks 10–15; 10-week term = Weeks 5–10



8. Risks and Resources



Risk 1: The Kaggle dataset may not have the bounding-box annotations needed for detection.

Plan B: Convert/create the needed labels or select another free public detection dataset.

Risk 2: Model training may not produce good license plate detections.

Plan B: Use a pretrained YOLOv8 model and reduce the project scope if needed.

Compute: Google Colab or Kaggle free resources.

Cost: \$0; planned models, datasets, frameworks, and tools will be free tier or open source.